Outpatient Clinic

This report presents a data analysis for a fictional clinic. The data analysis procedure will be checking the appointments, patients' area, specialties, minimum & maximum ages, lead times between the bookings dates & appoin1tments dates and the attendance status. The analysis covers the period from January 1 2024 to December 31 2025. The two tables provided are the: *appointments.csv* containing 5.015 rows and the *patients.csv* containing 1.005 rows. All data are synthetic and for educational purposes.

The project is structured into three stages:

1. Data quality check and cleaning: identifying data quality issues and preparing the datasets for analysis.
2. Data analysis: exploring the cleaned data to examine appointment attendance and identify patterns and trends.
3. Findings and recommendations: addressing stakeholders' questions, explaining the key insights, and presenting recommendations supported by the analysis.

Data Quality

(*appointments.csv* & *patients.csv*)

DQ_001

Check for nulls in the appointment_id column. The query returned no rows.

Query:

```
SELECT
  appointment_id
FROM
  `Outpatient_Clinic.appointments`
WHERE
  appointment_id IS NULL
```

DQ_002

Identify unique records with the distinct command.

Query:

```
SELECT
  DISTINCT appointment_id
FROM
  `Outpatient_Clinic.appointments`
```

Results:
Query returned 5.000 rows.

DQ_003

Check for the exact duplicate rows by their appointment_id and how many times they appear in the table.

Query:

```
SELECT
  appointment_id, count(*) AS duplicate_app_id
FROM
  `Outpatient_Clinic.appointments`
GROUP BY
  appointment_id
HAVING
  duplicate_app_id > 1
```

Results:

Row	appointment_id	duplicate_app_id
1	A02280	2
2	A03613	2
3	A02498	2
4	A03460	2
5	A03741	2
6	A01974	2
7	A01355	2
8	A04926	2
9	A01990	2
10	A02183	2
11	A00337	2
12	A03857	2
13	A01088	2
14	A00944	2
15	A00632	2

DQ_004

Check for inconsistencies in the attendance_status column. No inconsistencies were found.

Query:

```
SELECT
  DISTINCT attendance_status
FROM
  `Outpatient_Clinic.appointments`
```

Results:

Row	attendance_status
1	No-show
2	Cancelled
3	Attended

DQ_005

Check for logical inconsistency where there is a visit_duration_minutes and also the attendance_status is not Attended. Check passed - no results.

Query:

```
SELECT
  appointment_id, attendance_status, visit_duration_minutes
FROM
  `Outpatient_Clinic.appointments`
WHERE
  visit_duration_minutes IS NOT NULL AND (attendance_status = "No-show"
or attendance_status = "Cancelled")
```

DQ_006

Check for inconsistencies in the specialty column. Inconsistencies found.

Query:

```
SELECT
  DISTINCT specialty
FROM
  `Outpatient_Clinic.appointments`
```

Results:

Row	specialty
1	GENERAL MEDICINE
2	General Medicine
3	Pulmonology
4	PULMONOLOGY
5	Orthopedics
6	ENDOCRINOLOGY
7	ORTHOPEDICS
8	CARDIOLOGY
9	Cardiology
10	Cardiology
11	Endocrinology
12	General Medicine
13	Orthopedics
14	Pulmonology

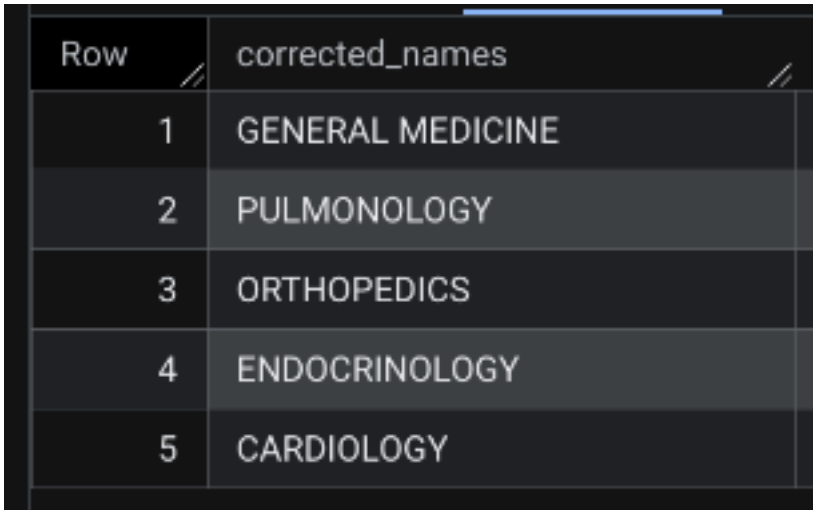
DQ_007

Check of how many specialties are in the table.

Query:

```
SELECT
  DISTINCT UPPER(TRIM(specialty)) AS corrected_names
FROM
  `Outpatient_Clinic.appointments`
```

Results:



Row	corrected_names
1	GENERAL MEDICINE
2	PULMONOLOGY
3	ORTHOPEDICS
4	ENDOCRINOLOGY
5	CARDIOLOGY

DQ_008

Check for NULLs in the booking_date column. No results.

Query:

```
SELECT
  booking_date
FROM
  `Outpatient_Clinic.appointments`
WHERE
  booking_date IS NULL
```

DQ_009

Check for NULLs in the appointment_date column. No results.

Query:

```
SELECT
  appointment_date
FROM
  `Outpatient_Clinic.appointments`
WHERE
  appointment_date IS NULL
```

DQ_010

Check for logical inconsistency where the booking_date is later in the calendar than the appointment_date. No results.

Query:

```
SELECT
  appointment_date, booking_date
FROM
  `Outpatient_Clinic.appointments`
WHERE
  booking_date > appointment_date
```

DQ_011

Checking for negative values in the visit_duration_minutes column. Identified 10 appointments with non-positive visit durations: nine negative values and one zero.

Query:

```
SELECT
  appointment_id, visit_duration_minutes
FROM
  `Outpatient_Clinic.appointments`
WHERE
  visit_duration_minutes <= 0
```

Results:

Row	appointment_id	visit_duration_mi...
1	A01080	-10
2	A03172	-5
3	A01391	-10
4	A04131	-10
5	A04284	-5
6	A03816	0
7	A01498	-10
8	A00643	-5
9	A02273	-10
10	A04411	-10

DQ_012

Check for NULLs in the patient_id column. No results.

Query:

```
SELECT
  patient_id
FROM
  `Outpatient_Clinic.appointments`
WHERE
  patient_id IS NULL
```

DQ_013

Check for logical inconsistency between the visit_duration_minutes & attendance_status column. No results.

Query:

```
SELECT
  visit_duration_minutes
FROM
  `Outpatient_Clinic.appointments`
WHERE
  visit_duration_minutes IS NULL AND attendance_status = "Attended"
```

DQ_014

Check for logical inconsistency between appointment_date (appointments.csv) and date_of_birth (patients.csv). No results.

Query:

```
SELECT
  a.appointment_id,
  a.patient_id,
  p.date_of_birth,
  a.appointment_date
FROM
  `Outpatient_Clinic.appointments` AS a
INNER JOIN
  `Outpatient_Clinic.patients` AS p
ON
  a.patient_id = p.patient_id
WHERE
  p.date_of_birth > a.appointment_date
```

DQ_015

Check for appointments with patient_id, with that patient_id is not in the patients table. No results.

Query:

```
SELECT
  a.patient_id,
  a.appointment_id
FROM
  `Outpatient_Clinic.appointments_clean` AS a
LEFT JOIN
  `Outpatient_Clinic.patients_clean` AS p
ON a.patient_id = p.patient_id
WHERE
  p.patient_id IS NULL
```

DQ_001_patients

Check for NULLs in the patient_id column. No results.

Query:

```
SELECT
  patient_id
FROM
  `Outpatient_Clinic.patients`
WHERE
  patient_id IS NULL
```

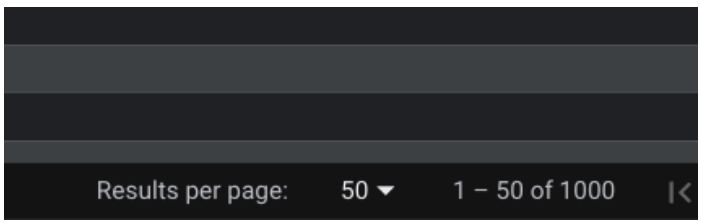
DQ_002_patients

Check for the distinct number of patient_ids.

Query:

```
SELECT
  DISTINCT patient_id
FROM
  `Outpatient_Clinic.patients`
```

Results:
1.000 rows.



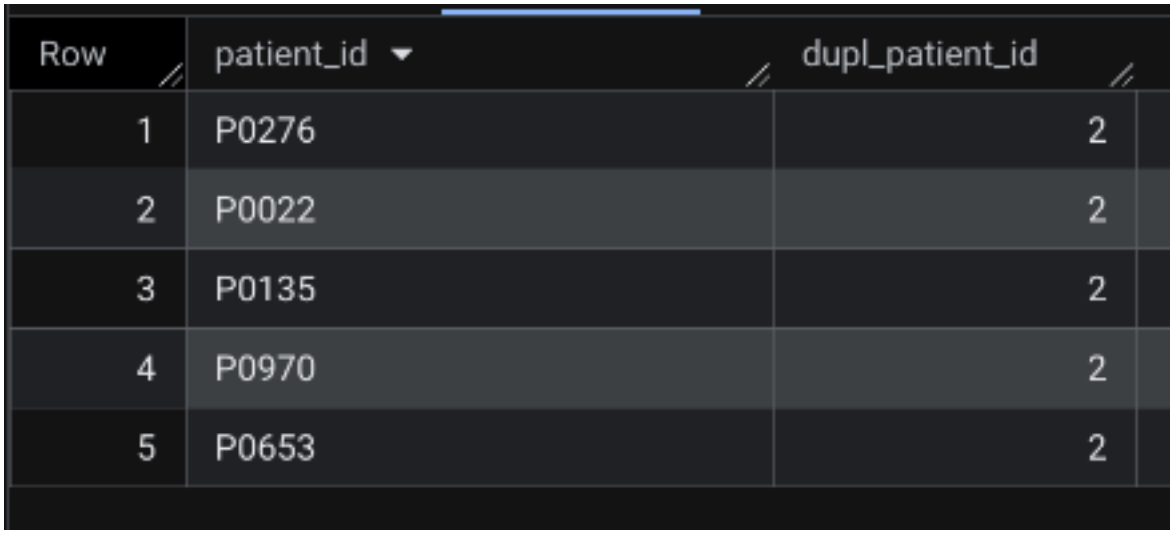
DQ_003_patients

Check for duplicates in the patient_id column.

Query:

```
SELECT
  patient_id, COUNT(*) AS dupl_patient_id
FROM
  `Outpatient_Clinic.patients`
GROUP BY
  patient_id
HAVING
  dupl_patient_id > 1
```

Results:



Row	patient_id	dupl_patient_id
1	P0276	2
2	P0022	2
3	P0135	2
4	P0970	2
5	P0653	2

DQ_004_patients

Check for consistency in area column.

Query:

```
SELECT
  DISTINCT area
FROM
  `Outpatient_Clinic.patients`
```

Results:

Row	area
1	Central
2	East
3	North
4	South
5	West
6	South
7	central
8	Central
9	south
10	East
11	east
12	north

DQ_005_patients

Check for the correct values in the area column.

Query:

```
SELECT
  DISTINCT UPPER(TRIM(area)) AS corrected_area
FROM
  `Outpatient_Clinic.patients`
```

Results:

Row	corrected_area
1	CENTRAL
2	EAST
3	NORTH
4	SOUTH
5	WEST

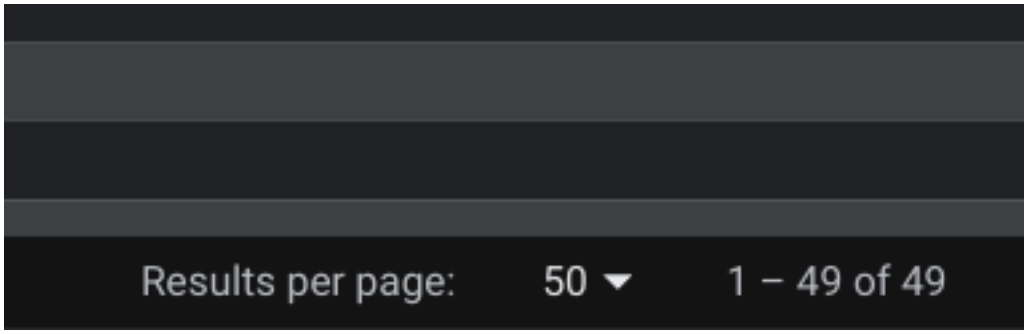
DQ_006_patients

Check for NULLs in the column chronic_condition_category. 49 rows found.

Query:

```
SELECT
  chronic_condition_category
FROM
  `Outpatient_Clinic.patients`
WHERE
  chronic_condition_category IS NULL
```

Results:



DQ_007_patients

Check for inconsistencies in the column chronic_condition_category.

Query:

```
SELECT
  DISTINCT chronic_condition_category
FROM
  `Outpatient_Clinic.patients`
```

Results:

Row	chronic_condition_category
1	*null*
2	Cardiovascular
3	Diabetes
4	Musculoskeletal
5	None
6	Other
7	Respiratory

DQ_008_patients

Check for NULLs in the column date_of_birth. No results.

Query:

```
SELECT
  date_of_birth
FROM
  `Outpatient_Clinic.patients`
WHERE
  date_of_birth IS NULL
```

DQ_009_patients

Check for minimum & maximum dates in date_of_birth column.

Query:

```
SELECT
  MIN(date_of_birth) AS min_bod,
  MAX(date_of_birth) AS max_bod
FROM
  `Outpatient_Clinic.patients`
```

Results:

Row	min_bod	max_bod
1	1938-03-06	2020-12-26

DQ_010_patients

Check for patients without registered area. No results.

Query:

```
SELECT
  patient_id,
  area
FROM
  `Outpatient_Clinic.patients`
WHERE
  area IS NULL
  OR TRIM(area) = '';
```

Data Cleaning

(*appointments.csv* & *patients.csv*)

In this stage we will create a new clean table of the appointments table without the duplicates. Also there will be some alterations in the new table by adding a column that will help us later in the analysis.

DC_000

Creation of a new table from the table appointments without the duplicates. New table contains 5.000 rows.

Query:

```
CREATE TABLE
    `Outpatient_Clinic.appointments_clean` AS
SELECT
    DISTINCT *
FROM
    `Outpatient_Clinic.appointments`
```

DC_001

Update the column specialy for the inconsistency with the characters (upper, lower and spaces in the records).

Query:

```
UPDATE
  `Outpatient_Clinic.appointments_clean`
SET
  specialty = UPPER(TRIM(specialty))
WHERE
  specialty <> UPPER(TRIM(specialty))
```

DC_002

Update the column visit_duration_minutes. All the negative values will become positive and all the values that are 0 will become null. Everything else stays as it is. (There was confirmation from the source about these changes.)

Query:

```
UPDATE
  `Outpatient_Clinic.appointments_clean`
SET
  visit_duration_minutes =
  CASE
    WHEN visit_duration_minutes < 0 THEN ABS(visit_duration_minutes)
    WHEN visit_duration_minutes = 0 THEN NULL
    ELSE  visit_duration_minutes
  END
WHERE
```

```
  visit_duration_minutes <= 0
```

DC_003

Alteration of the table by adding one new column. This column will help us later in the analysis. The new column lead_time_days will represent the days between the appointment_date and the booking_date.

Query:

```
ALTER TABLE
  `Outpatient_Clinic.appointments_clean`
ADD COLUMN
  lead_time_days INT64;
UPDATE
  `Outpatient_Clinic.appointments_clean`
SET
  lead_time_days = DATE_DIFF(appointment_date, booking_date, DAY)
WHERE
  TRUE;
```

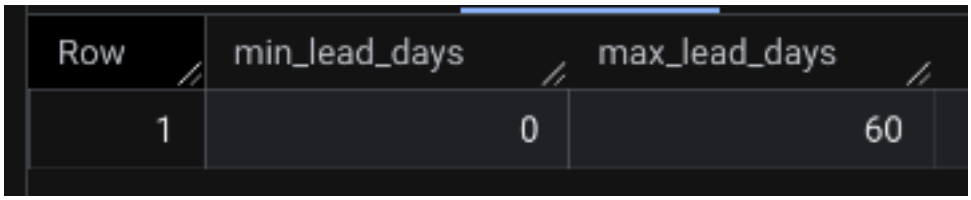
DC_004

Checking the range (min & max) of the new column - lead_time_days.

Query:

```
SELECT
  MIN(lead_time_days) AS min_lead_days,
  MAX(lead_time_days) AS max_lead_days
FROM
  `Outpatient_Clinic.appointments_clean`
```

Results:



Row	min_lead_days	max_lead_days
1	0	60

DC_005

Alteration of the table by adding a new column with the name no_show_flag. Update The newly created column for us to show in numbers when an appointment has a flag of no-show and attended by numbers or cancelled by nulls.

Query:

```
ALTER TABLE
  `Outpatient_Clinic.appointments_clean`
ADD COLUMN
  no_show_flag INT64;

UPDATE
  `Outpatient_Clinic.appointments_clean`
SET
  no_show_flag =
    CASE
      WHEN attendance_status = "No-show" THEN 1
      WHEN attendance_status = "Attended" THEN 0
      WHEN attendance_status = "Cancelled" THEN NULL
    END
WHERE
  TRUE
```

DC_006

Alteration of the table by adding a new column with the name age. We updated this new column for us to show the age of patients.

Query:

```
ALTER TABLE
  `Outpatient_Clinic.appointments_clean`
ADD COLUMN
  age INT64;

UPDATE
   `Outpatient_Clinic.appointments_clean` AS a
SET age =
  DATE_DIFF(a.appointment_date, p.date_of_birth, YEAR)
  FROM
    `Outpatient_Clinic.patients_clean` AS p
  WHERE
    a.patient_id = p.patient_id;
```

DC_007

Alteration of the table by adding a new column with the name age_group. Updating this new column for us to divide the ages in 5 groups:

- 0-17
- 18-34
- 35-49
- 50-64
- 65plus

Query:

```
ALTER TABLE
  `Outpatient_Clinic.appointments_clean`
ADD COLUMN
  age_group STRING;

UPDATE
  `Outpatient_Clinic.appointments_clean`
SET
  age_group =
    CASE
      WHEN age BETWEEN 0 and 17 THEN "0-17"
      WHEN age BETWEEN 18 AND 34 THEN "18-34"
      WHEN age BETWEEN 35 AND 49 THEN "35-49"
      WHEN age BETWEEN 50 AND 64 THEN "50-64"
      WHEN age > 64 THEN "65plus"
    END
WHERE
  TRUE;
```

DC_008

Checking the minimum and maximum ages grouped by the *age_group.*

Query:

```
SELECT
  age_group,
  MIN(age) AS min_age,
  MAX(age) AS max_age,
  COUNT(*) AS appointment_count
FROM
  `Outpatient_Clinic.appointments_clean`
GROUP BY
  age_group
ORDER BY
  appointment_count DESC
```

Results:

Row	age_group	min_age	max_age	appointment_count
1	65plus	65	87	1384
2	0-17	4	17	960
3	35-49	35	49	956
4	50-64	50	64	871
5	18-34	18	34	829

DC_001_patients

Create a new table from patients table without any duplicates. New table will contain 1.000 rows.

Query:

```
CREATE TABLE
   `Outpatient_Clinic.patients_clean` AS
SELECT
   DISTINCT *
FROM
   `Outpatient_Clinic.patients`;
```

Data Analysis

(*appointments.csv* & *patients.csv*)

This is the second stage of the whole process. In this stage we will explain all the findings and we will answer the questions given by the stakeholders. Some answers will contain graphs to make the findings easier to understand.

Question 1:

How many patients live in each area, and what percentage of all registered patients does each area represent?

Answer (DA_001):

Query:

```
SELECT
  area,
  COUNT(*) AS patients_num,
  (COUNT(*)*100/1000) AS patients_pct
FROM
  `Outpatient_Clinic.patients_clean`
GROUP BY
  area
ORDER BY
  patients_num DESC
```

Results:

Row	area	patients_num	patients_pct
1	NORTH	216	21.6
2	CENTRAL	205	20.5
3	WEST	198	19.8
4	SOUTH	191	19.1
5	EAST	190	19.0

Findings:

North area has the most patients with 216 people and the percentage of them is 21,6%. In addition, the area with the fewest people is East with 190 (19%). In between we can see that the central has 205 people (20,5%), West with 198 (19,8%) and South with 191 (19,1%).

Question 2:

How many patients are in each chronic condition category, and what percentage of all registered patients does each category represent?

Answer (DA_002):

```
SELECT
  chronic_condition_category,
  COUNT(*) AS patients_num,
  (COUNT(*)*100/1000) AS patients_pct
FROM
  `Outpatient_Clinic.patients_clean`
GROUP BY
  chronic_condition_category
ORDER BY
  patients_num DESC
```

Results:

Row	chronic_condition_category	patients_num	patients_pct
1	None	469	46.9
2	Musculoskeletal	109	10.9
3	Other	104	10.4
4	Respiratory	92	9.2
5	Cardiovascular	92	9.2
6	Diabetes	86	8.6
7	*null*	48	4.8

Findings:

The most common category of chronic_condition_category is None. 469 people (46,9%) no chronic condition recorded. In the second place there are 109 people (10,9%) with Musculoskeletal condition, Other condition with 104 (10,4%) and the following are:

Respiratory and Cardiovascular with 92 people each - (9,2%)
Diabetes with 86 people (8,6%) and 48 (4,8%) people with not registered values.

Question 3:

How many appointments are Attended, No-show, or Cancelled, and what percentage of all appointments does each status represent?

Answer (DA_003):

Query:

```
SELECT
  attendance_status,
  COUNT(*) AS appointments_num,
  (COUNT(*)*100/5000) AS appointments_pct
FROM
  `Outpatient_Clinic.appointments_clean`
GROUP BY
  attendance_status
ORDER BY
  appointments_num DESC
```

Results:

Row	attendance_status	appointments_num	appointments_pct
1	Attended	3873	77.46
2	No-show	614	12.28
3	Cancelled	513	10.26

Findings:

As we can see from the results, Attended status is the most popular with 3.873 records (77,46%). The following group is No-show with 614 records (12,28%) and in the third place we have the cancelled group with 513 records (10,26%).

Question 4:

For each year and month, how many appointments were scheduled, how many were Attended, No-show, or Cancelled, and what was the no-show rate?

Answer(DA_004):

Query:

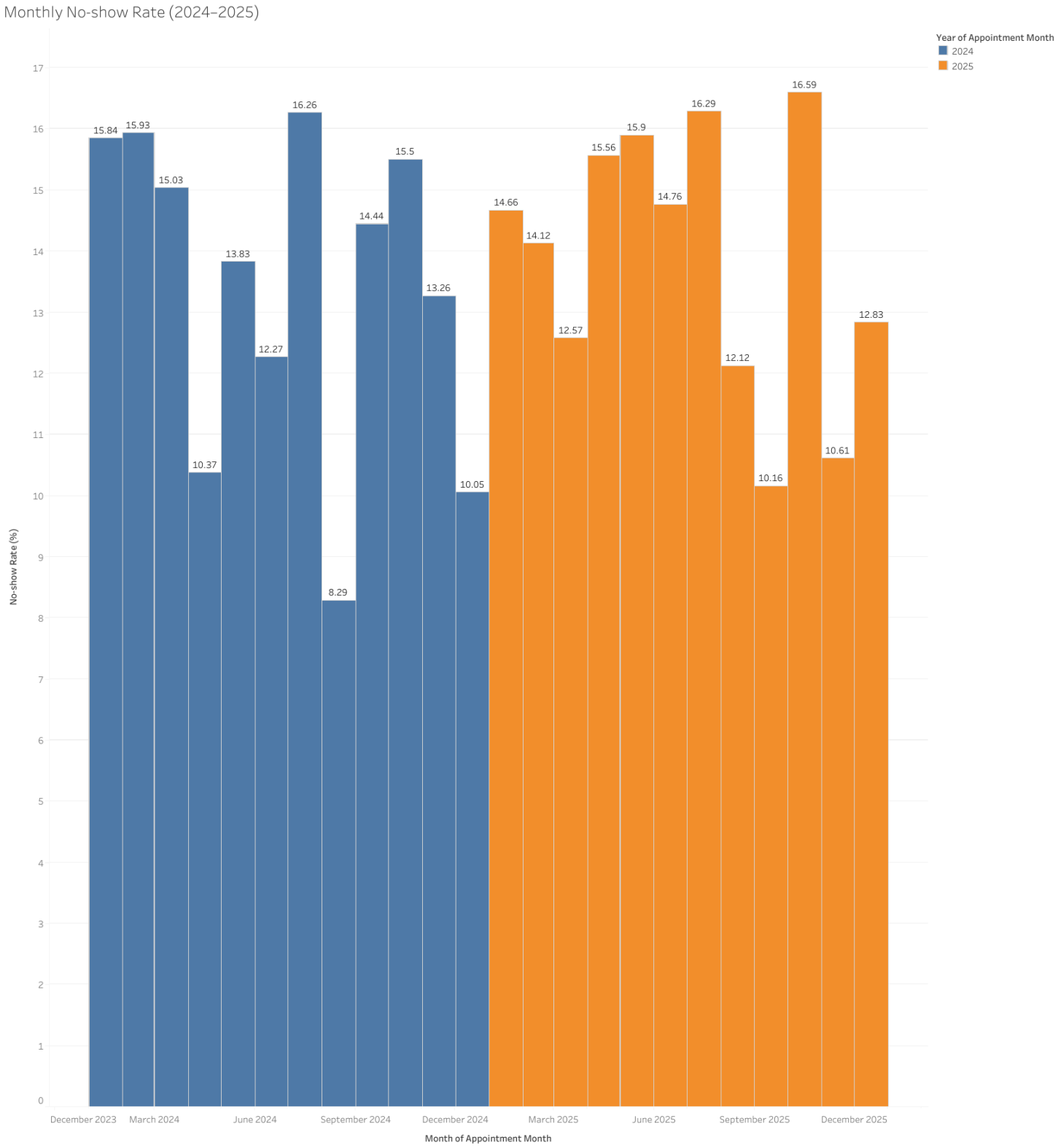
```
SELECT
  EXTRACT(YEAR FROM appointment_date) AS app_year,
  EXTRACT(MONTH FROM appointment_date) AS app_month,
  COUNT(*) AS appointments_num,
  COUNTIF(attendance_status = "Attended") AS attended_num,
  COUNTIF(attendance_status = 'No-show') AS no_show_num,
  COUNTIF(attendance_status = 'Cancelled') AS cancelled_num,
  ROUND(AVG(no_show_flag) * 100, 2) AS no_show_rate
FROM
  `Outpatient_Clinic.appointments_clean`
GROUP BY
  app_year,
  app_month
ORDER BY
  app_year,
  app_month
```

Results:

Row	app_year	app_month	appointments_num	attended_num	no_show_num	cancelled_num	no_show_rate
1	2024	1	222	170	32	20	15.84
2	2024	2	201	153	29	19	15.93
3	2024	3	212	164	29	19	15.03
4	2024	4	194	147	17	30	10.37
5	2024	5	210	162	26	22	13.83
6	2024	6	174	143	20	11	12.27
7	2024	7	222	170	33	19	16.26
8	2024	8	225	188	17	20	8.29
9	2024	9	215	160	27	28	14.44
10	2024	10	229	169	31	29	15.5
11	2024	11	204	157	24	23	13.26
12	2024	12	216	170	19	27	10.05
13	2025	1	218	163	28	27	14.66
14	2025	2	182	146	24	12	14.12
15	2025	3	204	160	23	21	12.57
16	2025	4	204	152	28	24	15.56
17	2025	5	221	164	31	26	15.9
18	2025	6	237	179	31	27	14.76
19	2025	7	201	149	29	23	16.29
20	2025	8	182	145	20	17	12.12
21	2025	9	200	168	19	13	10.16
22	2025	10	223	171	34	18	16.59
23	2025	11	202	160	19	23	10.61
24	2025	12	202	163	24	15	12.83

Findings:

In the above results table we can see and check all the appointments by months and year. Furthermore there are columns that are showing all the attendance_status stages with numbers and percentages. The highest no-show rate was in October 2025 at 16.59%, while the lowest was in August 2024 at 8.29%.



Question 5:

For each specialty, how many appointments were scheduled, how many were Attended, No-show, or Cancelled, and what was the no-show rate?

Answer(DA_005):

Query:

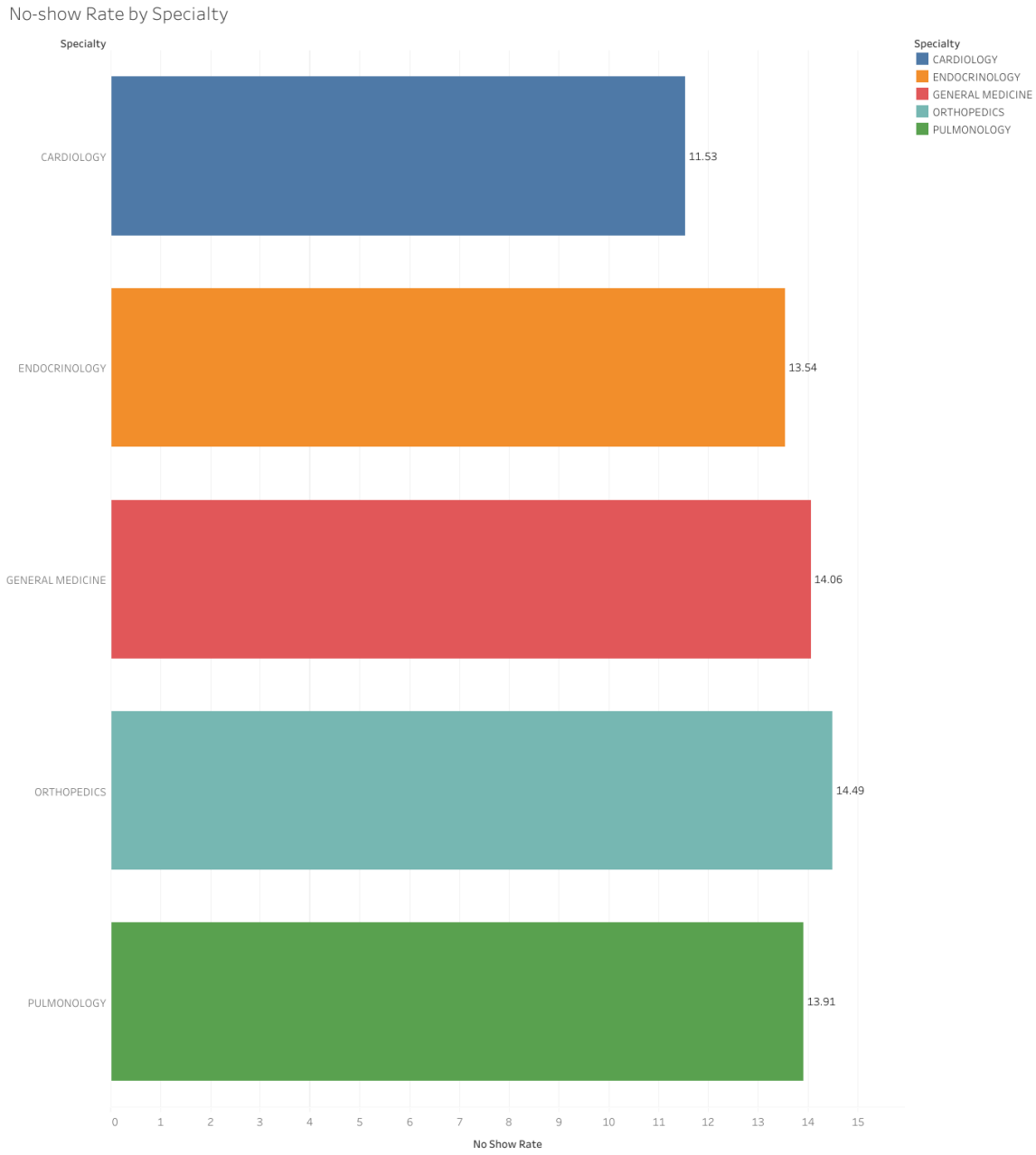
```
SELECT
  specialty,
  COUNT(*) AS appointments_num,
  COUNTIF(attendance_status = "Attended") AS attended_num,
  COUNTIF(attendance_status = "No-show") AS no_show_num,
  COUNTIF(attendance_status = "Cancelled") AS cancelled_num,
  ROUND(AVG(no_show_flag)*100, 2) AS no_show_rate
FROM
  `Outpatient_Clinic.appointments_clean`
GROUP BY
  specialty
ORDER BY
  appointments_num DESC
```

Results:

Row	specialty	appointments_num	attended_num	no_show_num	cancelled_num	no_show_rate
1	GENERAL MEDICINE	2137	1638	268	231	14.06
2	ORTHOPEDICS	755	590	100	65	14.49
3	PULMONOLOGY	723	551	89	83	13.91
4	ENDOCRINOLOGY	708	549	86	73	13.54
5	CARDIOLOGY	677	545	71	61	11.53

Findings:

The specialty with the most appointments is GENERAL MEDICINE with 2137 appointments and CARDIOLOGY has the fewest appointments. Orthopedics had the highest no-show rate at 14.49%, while Cardiology had the lowest at 11.53%.


No-show Rate by Specialty
Specialty
CARDIOLOGY
ENDOCRINOLOGY
GENERAL MEDICINE
ORTHOPEDICS
PULMONOLOGY
11.53
13.54
14.06
14.49
13.91
0
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
No Show Rate

Question 6:

For each specialty, how many Attended visits had a duration greater than zero minutes, and what was their average duration in minutes?

Answer(DA_006):

Query:

```
SELECT
  specialty,
  COUNT(*) AS attended_num,
  ROUND(AVG(visit_duration_minutes), 2) AS avg_visit_time
FROM
  `Outpatient_Clinic.appointments_clean`
WHERE
  attendance_status = "Attended"
  AND visit_duration_minutes > 0
GROUP BY
  specialty
ORDER BY
  attended_num DESC
```

Results:

Row	specialty	attended_num	avg_visit_time
1	GENERAL MEDICINE	1637	22.54
2	ORTHOPEDICS	590	22.08
3	PULMONOLOGY	551	22.54
4	ENDOCRINOLOGY	549	21.97
5	CARDIOLOGY	545	27.26

Findings:

GENERAL MEDICINE has the most attended appointments (1637) with an average duration of 22,54 minutes. The group with the least appointments number is CARDIOLOGY with 545 appointments but with the longest average duration at 27,26 minutes.

Question 7:

How many unique patients attended at least one appointment in each specialty?

Answer(DA_007):

Query:

```
SELECT
  specialty,
  COUNT(DISTINCT patient_id) AS unique_patients
FROM
  `Outpatient_Clinic.appointments_clean`
WHERE
  attendance_status = "Attended"
GROUP BY
  specialty
ORDER BY
  unique_patients DESC
```

Results:

Row	specialty	unique_patients
1	GENERAL MEDICINE	639
2	ORTHOPEDICS	391
3	PULMONOLOGY	384
4	CARDIOLOGY	349
5	ENDOCRINOLOGY	348

Findings:

General Medicine served the highest number of unique patients 639. Followed by ORTHOPEDICS with 391, PULMONOLOGY with 384, CARDIOLOGY with 349 and ENDOCRINOLOGY with 348.

Question 8:

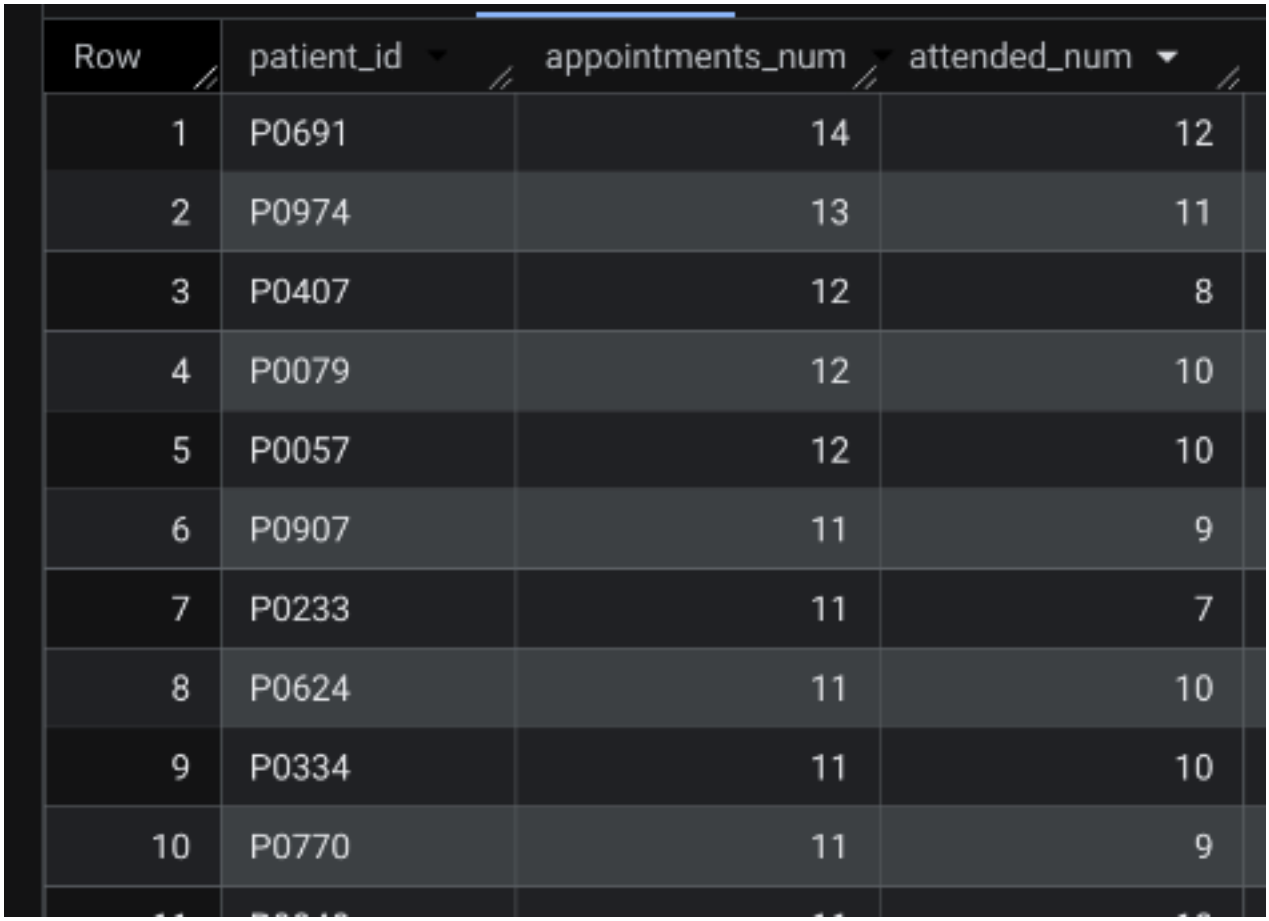
How many total appointments and Attended visits did each registered patient have, including patients with no appointments?

Answer(DA_008):

Query:

```
SELECT
  p.patient_id,
  COUNT(a.appointment_id) AS appointments_num,
  COUNTIF(a.attendance_status = "Attended") AS attended_num
FROM
  `Outpatient_Clinic.patients_clean` AS p
  LEFT JOIN
  `Outpatient_Clinic.appointments_clean` AS a
ON
  p.patient_id = a.patient_id
GROUP BY
    p.patient_id
ORDER BY
  appointments_num DESC
```

Results:



Row	patient_id	appointments_num	attended_num
1	P0691	14	12
2	P0974	13	11
3	P0407	12	8
4	P0079	12	10
5	P0057	12	10
6	P0907	11	9
7	P0233	11	7
8	P0624	11	10
9	P0334	11	10
10	P0770	11	9

Findings:

In a table of 1.000 rows of results, we can see that the patient with patient_id P0691 has the highest number of appointments with 14 appointments and attended 12 of them.

Question 9:

For each area, how many patients were registered, how many had no appointments, and what percentage of that area's patients had no appointments?

Answer(DA_009):

Query:

```
SELECT
  p.area,
  COUNT(DISTINCT p.patient_id) AS patients_num,
  COUNTIF(a.appointment_id IS NULL) AS no_appointment,
  ROUND(COUNTIF(a.appointment_id IS NULL) * 100.0 / COUNT(DISTINCT
p.patient_id), 2) AS no_appointment_pct
FROM
  `Outpatient_Clinic.patients_clean` AS p
LEFT JOIN
  `Outpatient_Clinic.appointments_clean` AS a
ON
  p.patient_id = a.patient_id
GROUP BY
  p.area
ORDER BY
  no_appointment_pct DESC
```

Results:

Row	area	patients_num	no_appointment	no_appointment_pct
1	SOUTH	191	25	13.09
2	WEST	198	21	10.61
3	NORTH	216	22	10.19
4	CENTRAL	205	19	9.27
5	EAST	190	13	6.84

Findings:

The area with the most registered patients that have no appointments at all is SOUTH with 25 no-appointments (13,09%). The area with the least no-appointmets is EAST with a number of 13 no-appointments (6,84%).

Question 10:

For each specialty, how many appointments were scheduled, including Cancelled appointments, and what was the average number of days between booking and the appointment date?

Answer(DA_010):

Query:

```
SELECT
  specialty,
  COUNT(*) AS appointments_num,
  ROUND(AVG(lead_time_days),2) AS avg_lead_time
FROM
  `Outpatient_Clinic.appointments_clean`
GROUP BY
  specialty
ORDER BY
  appointments_num DESC
```

Results:

Row	specialty	appointments_num	avg_lead_time
1	GENERAL MEDICINE	2137	24.74
2	ORTHOPEDICS	755	25.09
3	PULMONOLOGY	723	25.77
4	ENDOCRINOLOGY	708	25.39
5	CARDIOLOGY	677	23.26

Findings:

The specialty with the most appointments scheduled is GENERAL MEDICINE with 2.137 appointments and An average lead time of 24,74 days. The specialty with the longest average lead time is PULMONOLOGY with 723 appointments and an average lead time at 25,77 days.

Question 11:

For each day of the week, how many appointments were not cancelled, how many were No-show, and what was the no-show rate?

Answer(DA_011):

Query:

```
SELECT
  EXTRACT(DAYOFWEEK FROM appointment_date) AS weekday,
  COUNT(*) AS appointment_num,
  COUNTIF(attendance_status = "No-show") AS no_show_num,
  ROUND(AVG(no_show_flag) * 100, 2) AS avg_no_show
FROM
  `Outpatient_Clinic.appointments_clean`
WHERE
  attendance_status <> "Cancelled"
GROUP BY
  weekday
ORDER BY
  weekday
```

Results:

Row	weekday	appointment_num	no_show_num ▼	avg_no_show
1	2	735	98	13.33
2	3	780	106	13.59
3	4	784	97	12.37
4	5	736	116	15.76
5	6	702	102	14.53
6	7	750	95	12.67

Findings:

Thursday had the highest no-show rate (15.76%), while Wednesday had the lowest (12.37%). Cancelled appointments were excluded.

Question 12:

For each booking lead-time group - 0–7, 8–14, 15–30, and 31+ days - how many appointments were not cancelled, how many were No-show, and what was the no-show rate?

Answer(DA_012):

Query:

```
SELECT
  CASE
    WHEN lead_time_days BETWEEN 0 AND 7 THEN "0-7"
    WHEN lead_time_days BETWEEN 8 AND 14 THEN "8-14"
    WHEN lead_time_days BETWEEN 15 AND 30 THEN "15-30"
    WHEN lead_time_days >= 31 THEN "31plus"
  END AS lead_time_group,
  COUNT(*) AS appointments_num,
  COUNTIF(attendance_status = "No-show" ) AS no_show_sum,
  ROUND(AVG(no_show_flag)*100, 2) AS avg_no_show
FROM
  `Outpatient_Clinic.appointments_clean`
WHERE
  attendance_status <> "Cancelled"
GROUP BY
  lead_time_group
ORDER BY
  no_show_sum DESC
```

Results:

Row	lead_time_group	appointments_num	no_show_sum	avg_no_show
1	31plus	1768	323	18.27
2	0-7	1343	128	9.53
3	15-30	937	114	12.17
4	8-14	439	49	11.16

Findings:

The lead_time group with the most appointments were not cancelled and most no_show appointments is the 31plus days group with 1768 and 323 appointments respectively. The average no_show percentage of this group is 18,27%.

Question 13:

For appointments with and without reminders, how many were not cancelled, how many were No-show, and what was the no-show rate? Show these results both overall and separately for each specialty.

Answer(DA_013_1):

Query:

```
SELECT
  reminder_sent,
  COUNT(*) AS appointments_num,
  COUNTIF(attendance_status = "No-show") AS no_show_num,
  ROUND(AVG(no_show_flag)*100, 2) AS avg_no_show_flag
FROM
  `Outpatient_Clinic.appointments_clean`
WHERE
  attendance_status <> "Cancelled"
GROUP BY
  reminder_sent
```

Results:

Row	reminder_sent	appointments_num	no_show_num	avg_no_show_flag
1	true	2960	371	12.53
2	false	1527	243	15.91

Findings:

From the results we can distinguish that appointments with reminders had a lower no-show rate (12,53%) than appointments with no reminders (15,91%). This suggests an association between reminders and better attendance, but does not establish that reminders caused the difference.

Answer(DA_013_2):

Query:

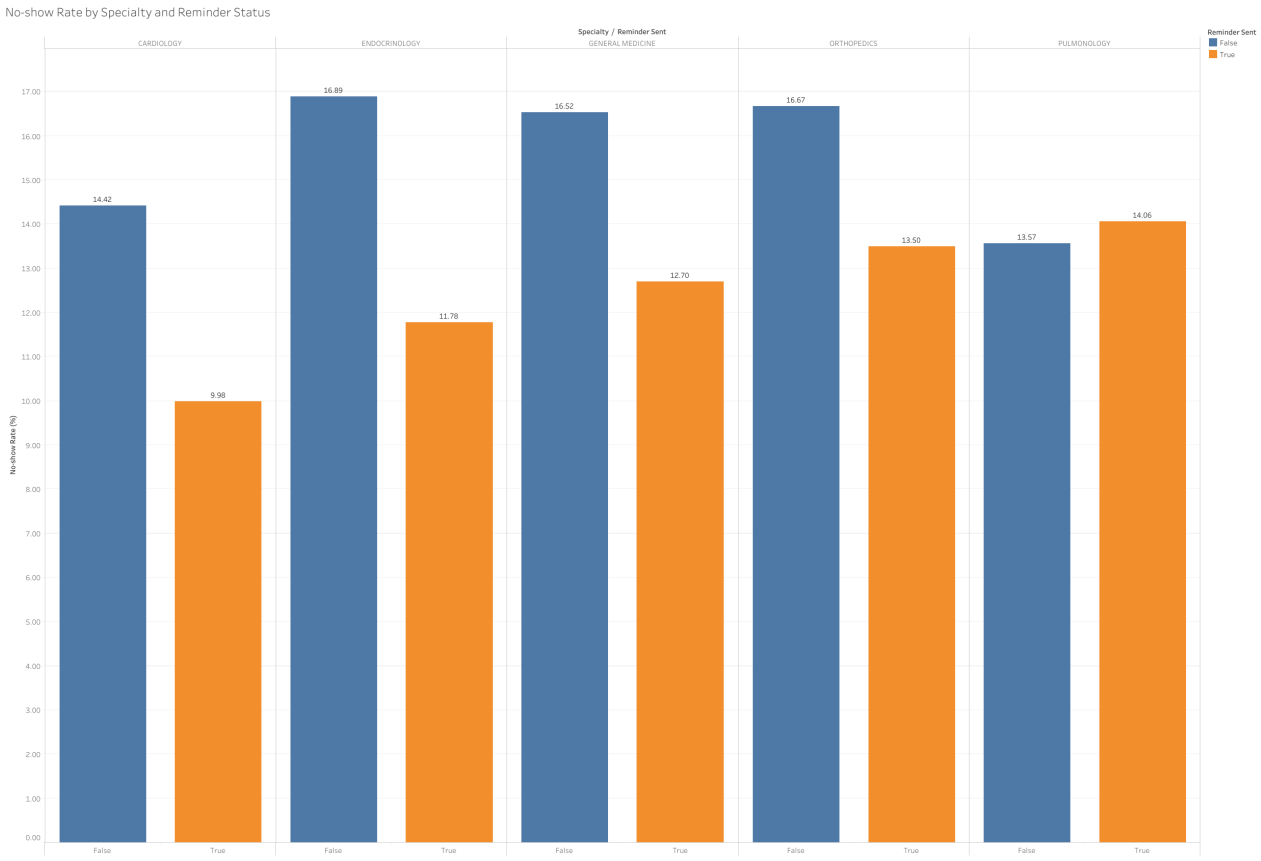
```
SELECT
  reminder_sent,
  specialty,
  COUNT(*) AS appointments_num,
  COUNTIF(attendance_status = "No-show") AS no_show_num,
  ROUND(AVG(no_show_flag)*100, 2) AS avg_no_show_flag
FROM
  `Outpatient_Clinic.appointments_clean`
WHERE
  attendance_status <> "Cancelled"
GROUP BY
  reminder_sent, specialty
ORDER BY
  reminder_sent DESC
```

Results:

Row	reminder_sent	specialty	appointments_num	no_show_num	avg_no_show_flag
1	true	GENERAL MEDICINE	1228	156	12.7
2	true	PULMONOLOGY	441	62	14.06
3	true	ENDOCRINOLOGY	416	49	11.78
4	true	ORTHOPEDICS	474	64	13.5
5	true	CARDIOLOGY	401	40	9.98
6	false	GENERAL MEDICINE	678	112	16.52
7	false	ENDOCRINOLOGY	219	37	16.89
8	false	ORTHOPEDICS	216	36	16.67
9	false	CARDIOLOGY	215	31	14.42
10	false	PULMONOLOGY	199	27	13.57

Findings:

Appointments with reminders had lower no-show rates in four of the five specialties. Endocrinology showed the largest difference (11.78% with reminders versus 16.89% without). Pulmonology was the exception, with a slightly higher rate among appointments with reminders (14.06% versus 13.57%). These results show an association, rather than evidence of causation.



Question 14:

For each age group, how many appointments were not cancelled, how many were No-show, and what was the no-show rate?

Answer(DA_014):

Query:

```
SELECT
  age_group,
  COUNT(*) AS appointments_sum,
  COUNTIF(attendance_status = "No-show") AS no_show_sum,
  ROUND(AVG(no_show_flag)*100, 2) AS avg_no_show_rate
FROM
`Outpatient_Clinic.appointments_clean`
WHERE
  attendance_status <> "Cancelled"
GROUP BY
  age_group
ORDER BY
age_group DESC
```

Results:

Row	age_group	appointments_sum	no_show_sum	avg_no_show_rate
1	65plus	1242	141	11.35
2	50-64	781	94	12.04
3	35-49	858	106	12.35
4	18-34	742	125	16.85
5	0-17	864	148	17.13

Findings:

No-show rates decreased across the age groups, from 17.13% among patients aged 0–17 to 11.35% among those aged 65 and over. Patients aged 18–34 also had a relatively high rate (16.85%), while the 35–49 and 50–64 groups recorded 12.35% and 12.04%, respectively. Cancelled appointments were excluded.

Conclusions, Recommendations & Limitations

This is the third and final stage of the analysis.

Conclusions

We checked 5,000 appointments and 1,000 patients. The total no-show rate is 13.68%, without the cancelled ones.General Medicine had the most appointments. But Orthopedics had the biggest no-show rate, 14.49%.
People who book appointment 31 days before or more, they don't come more often (18.27% no-show). Also young people miss appointments more. When we send reminders, no-show rate is lower (12.53%) than no reminders (15.91%). But this is not same for all specialties.

Recommendations

The clinic can send more reminders for appointments booked long time ago. Also make easy for patients to cancel or change time. Maybe check why young people and some specialties miss more.
If we check every month, we can see if things change and if attendance gets better.

Limitations

This project uses fake data for study, it is not for real clinic.
The analysis shows some connections, but not prove that reminders or age cause the missed appointments. We didn't check other things, like bus/car problems or exact time of appointment.
Age is not 100% exact, we just calculate with year. Missing values for conditions we keep separate. Nine numbers for duration was negative so we make them positive because error. One zero duration we change to NULL and don't count in average time.